Health-Aware Block Scheduling for LNG Carrier Production

ISE 572 — Reinforcement Learning Meets Shipbuilding

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The Problem

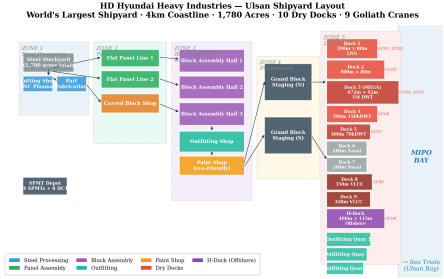
A $200M Scheduling Problem

An LNG carrier costs **$200M+** and takes **2–3 years** to build.

Each ship is assembled from **200 blocks**, each passing through **8 production stages**, moved by **24 transporters** and lifted by **9 cranes**.

Today, this scheduling is done **manually** — by experienced planners using spreadsheets and intuition.

Question: Can we learn a policy that schedules better than a human expert?



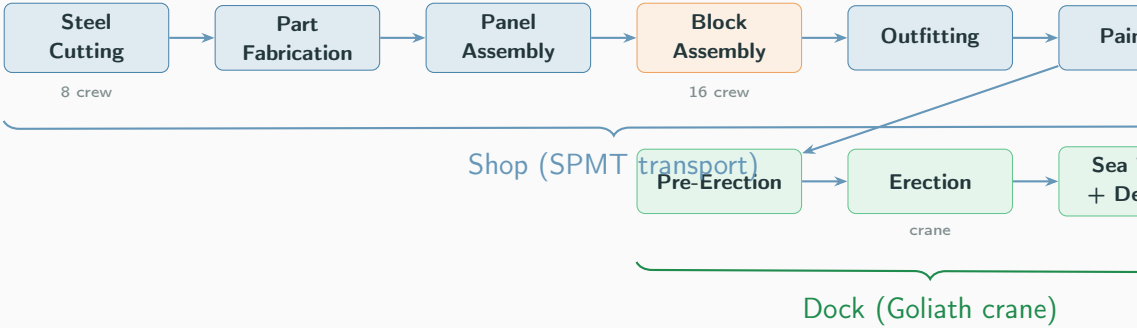
HD Hyundai Heavy Industries
Ulsan, South Korea

World's largest shipyard

What We Did

1. **Built a full simulation** of HHI Ulsan — 200 blocks, 10 docks, 9 cranes, 24 SPMTs, with NSRP-calibrated processing times
2. **Discovered entropy collapse** — PPO/SAC fail catastrophically in hierarchical masked action spaces (entropy $\rightarrow 0$ by epoch 5)
3. **Solved it with DAgger** — imitation learning achieves **118% of expert** on 50-block instances, bypassing the exploration problem entirely
4. **Exposed a scalability wall** — all methods (RL, OR, imitation) fail to scale beyond 200 blocks; simple EDD dispatch remains king at scale

How a Ship Gets Built



200 blocks/ship

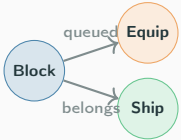
1.24M man-hours/ship

8 production stages

Method

Formulation: Scheduling as an MDP

State: Heterogeneous graph



GATv2 encoder $\rightarrow$ fixed $\mathbb{R}^{256}$ embedding

Works for any number of blocks

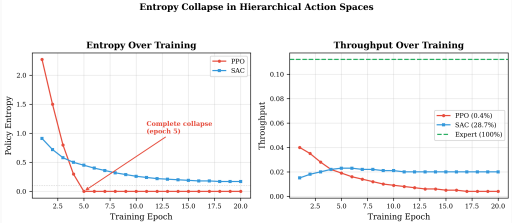
Action: Hierarchical (10 heads)

- *What?* transport / erect / hold
- *Which SPMT?* (1 of 24)
- *Which block?* (1 of 200)
- *Which crane?* (1 of 9)

Reward: $+1$ stage advance, $+10$ ship delivery, -0.5 per tardy step

Action masking ensures only valid actions are sampled — but this creates a problem...

The Core Problem: Entropy Collapse



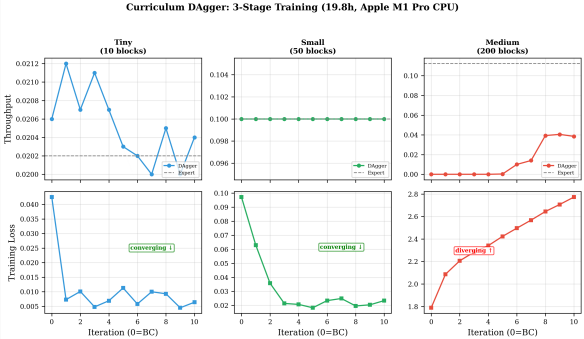
PPO entropy: **2.27** $\rightarrow$ **0.00** by epoch 5.

The policy becomes *completely deterministic* before it has learned anything useful.

Why? Action masking reduces valid options to 1–2 choices. Entropy is *mathematically bounded* by $\log k$, where k is the number of valid actions.

Result:
PPO: **0.4%** of expert

Our Solution: Learn from an Expert via DAgger



DAgger (Dataset Aggregation): Supervised learning from expert demonstrations — no exploration needed.

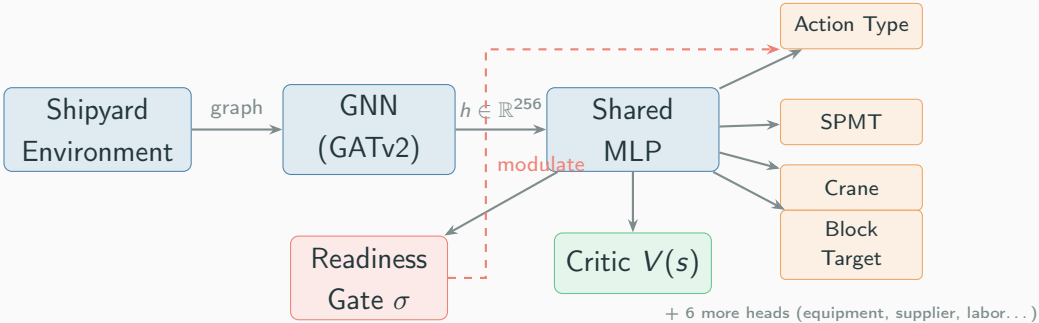
Curriculum: 3 stages

1. Tiny: 10 blocks
2. Small: 50 blocks
3. Medium: 200 blocks

Each stage: β anneals 0.8 $\rightarrow$ 0.2 (expert $\rightarrow$ student control)

Small instance: 118% of

Architecture



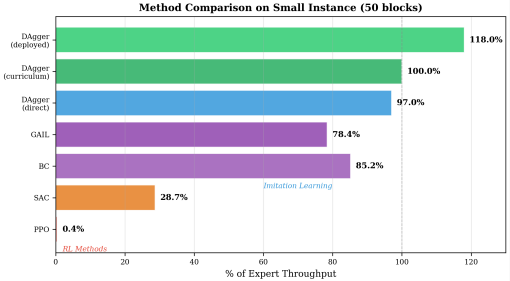
Readiness gate
logistic regression
modulates dispatch

10 action heads
only relevant heads
contribute to $\log \pi$

Action masking
invalid $\rightarrow -20$ logits
(soft, not $-\infty$)

Results

Result 1: DAgger Beats the Expert (on Small Instances)

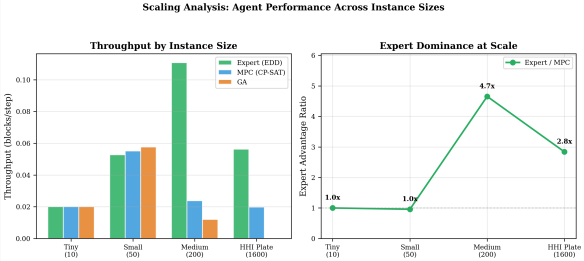


Method	vs Expert	
PPO	0.4%	✗
SAC	28.7%	✗
BC	85.2%	−
GAIL	78.4%	−
DAgger	**118%**	✓

DAgger *exceeds* expert by learning error recovery behaviors the rule-based EDD expert never exhibits.

10-seed validation: 97.0% ± 2.5%

Result 2: . . . But Nothing Scales to 200 Blocks



	Small (50 blocks)	Medium (200 blocks)
Expert	50 blocks	**200 blocks**
MPC	50 blocks	74 blocks

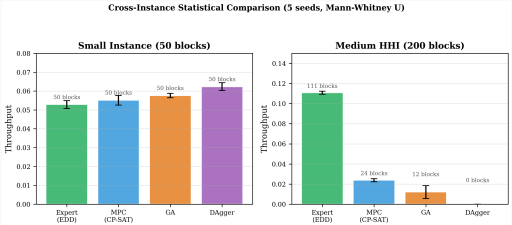
Under stochastic extensions:

Expert	50 blocks	**200 blocks**
MPC	50 blocks	70 blocks

Expert advantage grows with scale:

1.0× (small) → 2.7× (medium, det.) → 2.9× (stoch.)

Statistical Comparison (5 Seeds, Mann-Whitney U)



		Small (det / stoch)	Medium (det / stoch)
Expert	Blocks	50/50	200/200
	Thru.	0.052/0.049	0.100/0.100
MPC	Blocks	50/50	74/70
	Thru.	0.055/0.049	0.037/0.035

Medium Expert vs MPC: $p = 0.008$, $d \approx 27$

Stochastic: GA -13%, MPC -8%, Expert -4% (small)

Why Does DAgger Fail at Scale?

Curriculum training dynamics (medium stage):

Iter	β	Loss	Throughput
1	0.80	2.09	0.000
5	0.53	2.42	0.000
7	0.40	2.57	0.002
8	0.33	2.65	0.038
10	0.20	2.77	0.000

Loss *increases monotonically*.
The policy oscillates — learns transiently, then regresses at final checkpoint.

Root causes:

1. **Distribution shift**: 200-block state space has fundamentally different statistics than 50-block
2. **Capacity limit**: 256-dim MLP cannot represent the combinatorial 200-block action space
3. **Confirmed**: Retraining with normalizer persistence yields identical 0% result

Stochastic Simulation: Bridging the Sim-to-Real Gap

Five uncertainty sources modeled:

Extension	Distribution
Congestion	LogNormal(μ, 0.1)
Absences	Binomial(N, 0.05)
Fatigue	$\mathcal{N}(1, 0.05)$
Weather	4-state Markov
Duration	LogNormal, CV 5–25%

Combined per-step modifier:

$$\Delta t_{\text{eff}} = \Delta t \times p_{\text{shift}} \times m_{\text{weather}} \times f_{\text{dur}}$$

Impact on throughput:

	Expert	MPC	GA
Small (det)	0.053	0.055	0.059
Small (stoch)	0.050	0.050	0.051
Change	−4%	−8%	−13%
Med. (det)	0.100	0.037	—
Med. (stoch)	0.100	0.035	—
Change	**0%**	−7%	—

Key finding: GA most fragile (−13%);
Expert most robust;

Takeaways

Three Things to Remember

1

RL fails here

Entropy collapse in masked hierarchical action spaces is a *fundamental* problem, not a tuning issue. Validated across 3 domains.

2

Imitation works (at small scale)

DAgger achieves 118% of expert by learning error recovery. But it cannot generalize beyond its training distribution.

3

Simple rules scale best

EDD dispatch — a 1-line priority rule — outperforms MPC, GA, and learned policies at 200+ blocks. Scalability is the bottleneck.

Connection to ISE 572: Classic scheduling theory (EDD optimality for $1||\sum T_j$)

Limitations & Future Directions

What we couldn't solve

- 118% $\rightarrow$ 0% scalability cliff
- Entropy collapse (no RL fix found)
- Block assembly calibration ($R^2 = -0.03$)
- Simplified crane physics
- ~~No weather or disruptions~~ (now modeled)

What might work next

- **Attention-based heads** for variable entity counts (implemented, not yet trained)
- Larger policy networks (512+)
- Per-head adaptive entropy
- Transfer learning across ship types
- Digital twin integration

Demo

Live Demo: MES Dashboard

Live dashboard demo at `localhost:8050`

11 tabs: Overview · Blocks · Ships · Fleet · Health

Operations · Dependencies · GNN · Plates · Map · Docks

```
PYTHONPATH=src python experiments/live_simulation.py --policy expert
python -m src.mes.app
```

Questions?

Tests	214 passing
Instances	4 sizes (10–1600 blocks)
Agents	6 methods compared
Paper	36 pages
Dashboard	11 tabs, real-time
Training	50+ hours of experiments